

Incremental and Decremental Optimal Margin Distribution Learning

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Abstract

Incremental and decremental learning (IDL) deals with the tasks where new data arrives sequentially as a stream or old data turns unavailable continually due to the privacy protection. Existing IDL methods mainly focus on support vector machine and its variants with *linear*-type loss. There are few studies about the *quadratic*-type loss, whose Lagrange multipliers are unbounded and much more difficult to track. In this paper, we take the latest statistical learning framework *optimal margin distribution machine* (ODM) which involves a quadratic-type loss due to the optimization of margin variance, for example, and equip it with the ability to handle IDL tasks. Our proposed ID-ODM can avoid updating the Lagrange multipliers in an infinite range by determining their optimal values beforehand so as to enjoy much more efficiency. Moreover, ID-ODM is also applicable when multiple instances come and leave simultaneously. Extensive empirical studies show that ID-ODM can achieve $9.1\times$ speedup on average with almost no generalization lost compared to retraining ODM on new data set from scratch.

1 Introduction

For machine learning tasks, once data changes, the models should be refined. However, retraining models from scratch usually incurs unbearable cost or is even impossible in some situations, which gives rise to the *incremental and decremental learning* (IDL) [Schlimmer and Fisher, 1986; Cauwenberghs and Poggio, 2000]. This paradigm can efficiently update the model and be applied in many learning problems, just to name a few, anomaly detection [Laxhammar and Falkman, 2014; Nguyen *et al.*, 2019], concept drifting [Syed *et al.*, 1999; Gálmeanu and Andonie, 2022], active learning [Ertekin *et al.*, 2007; Huang *et al.*, 2020], privacy protection [Nguyen *et al.*, 2020; Sekhari *et al.*, 2021], and model estimation [Cherubin *et al.*, 2021].

Existing IDL methods mainly focus on *support vector machine* (SVM) [Cortes and Vapnik, 1995] and its variants.

We summarize the representative works of this line in Table 1. Based on the path following method, IDSVM [Cauwenberghs and Poggio, 2000] takes both incremental learning and decremental learning into account to update SVM. MID-SVM [Karasuyama and Takeuchi, 2009] is a batch version of IDSVM. AONSVM [Gu *et al.*, 2012] and FISVDD [Jiang *et al.*, 2019] extends the ν -SVM [Schölkopf *et al.*, 2000] and SVDD [Tax and Duin, 2004] to incremental learning, respectively. All these models use hinge loss. Besides, IL-S3VM [Gu *et al.*, 2018] and ILTSVM [Chen *et al.*, 2023] are incremental semi-supervised SVMs with symmetric hinge loss or its variant ramp loss.

On the one hand, the aforementioned methods mainly focus on large margin models, but the latest theoretical studies [Gao and Zhou, 2013] disclose that the margin distribution is much more critical to the generalization performance. On the other hand, these methods are limited to *linear*-type loss, in which the hinge loss is sensitive to outliers and symmetric hinge loss leads to non-convex optimization problems. In contrast, the *quadratic*-type loss is smoother and numerically easier to deal with. It results in a strictly diagonally dominant matrix in dual quadratic form which is always invertible, thus relaxes the restriction of *linear*-type loss that data should be linearly independent. As far as we know, there are few IDL methods concerning models with *quadratic*-type loss.

In this paper, we take the latest statistical learning framework *optimal margin distribution machine* (ODM) [Zhang and Zhou, 2019] which involves a *quadratic*-type loss due to the optimization of margin variance as an example and equip it with the ability to handle IDL tasks. The drawback of *quadratic*-type loss lies in the optimization aspect, that is the Lagrange multipliers are unbounded and much more difficult to track when data varies, and existing IDL methods with *linear*-type loss can hardly be adapted to these circumstances. To overcome this difficulty, we estimate the optimal value of the Lagrange multiplier of the varying data in advance to avoid the optimization in an infinite range, and then update along the fastest direction to make the interruption by breakpoints less frequent. We also theoretically analyze the time complexity and convergence of our proposed incremental and decremental ODM (ID-ODM). The remarkable advantages are as follows:

- It is the first attempt to implement IDL for kernel methods optimizing the margin distribution.

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